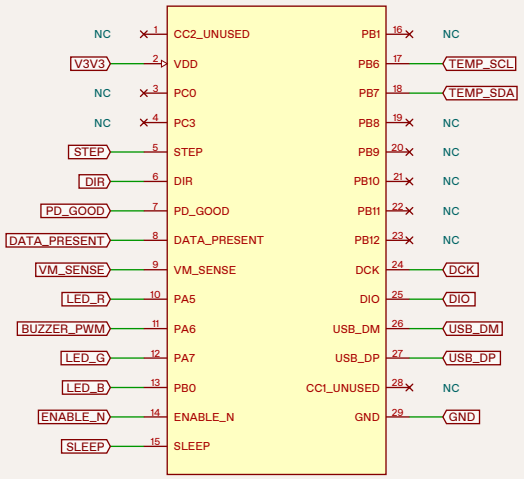
NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 1/13 - U_MCU

U_MCU



CH32X035G8U6

COMPONENT PURPOSE

U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
Firmware must check PD before motor enable.
Internal clock avoids an external crystal.

U_PD - CH224K

USB-C PD sink requests a 15 V motor supply.
PG reports a completed power negotiation.

U_BUCK - AP63203WU-7

3.3 V buck for MCU, logic and PWM buzzer.
Diode OR accepts programming USB or PD.

U_DRV - A4988

Two current-regulated bridges drive the coils.
STEP/DIR interface; fixed 1/16 microstepping.
0.24 ohm sense resistors set phase current.

U_ESD - USBLC6-2SC6

Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the USB protection diodes.

U_TEMP / BZ1 / D_STATUS / D_PWR

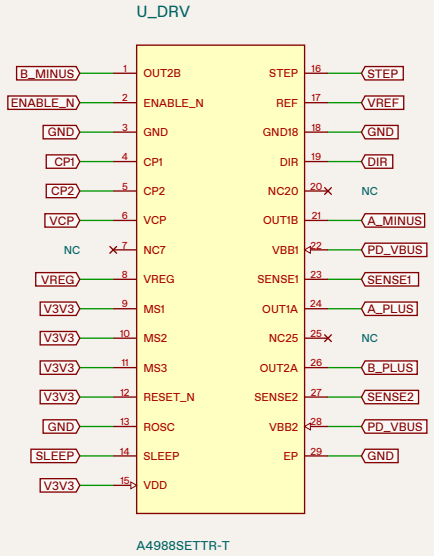
TMP102: I2C temperature, address 0x48.
PWM magnetic buzzer and RGB status need firmware.
Power LED indicates 3.3 V; RGB uses USB 5 V.

Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 2/13 - U_DRV



COMPONENT PURPOSE

U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
Firmware must check PD before motor enable.
Internal clock avoids an external crystal.

U_PD - CH224K

USB-C PD sink requests a 15 V motor supply.
PG reports a completed power negotiation.

U_BUCK - AP63203WU-7

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Diode OR accepts programming USB or PD.

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0.24 ohm sense resistors set phase current.

U_ESD - USBLC6-2SC6

Clamps ESD on USB D+ and D- at the data port.
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U_TEMP / BZ1 / D_STATUS / D_PWR

TMP102: I2C temperature, address 0x48.
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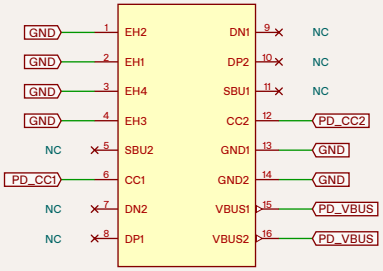
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Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

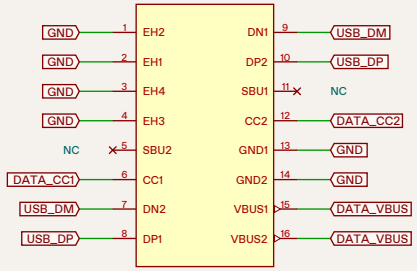
SHEET 3/13 - J_PD, J_DATA, U_PD, R_PD, C_PD, R_PG

J_PD



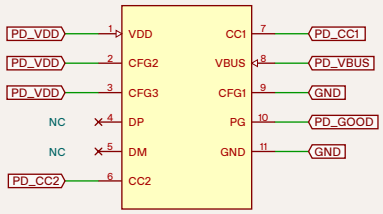
TYPE-C-31-M-12

J_DATA



TYPE-C-31-M-12

U_PD



CH224K







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PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

COMPONENT PURPOSE

U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
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Internal clock avoids an external crystal.

U_PD - CH224K

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PG reports a completed power negotiation.

U_BUCK - AP63203WU-7

3.3 V buck for MCU, logic and PWM buzzer.
Diode OR accepts programming USB or PD.

U_DRV - A4988

Two current-regulated bridges drive the coils.
STEP/DIR interface; fixed 1/16 microstepping.
0.24 ohm sense resistors set phase current.

U_ESD - USBLC6-2SC6

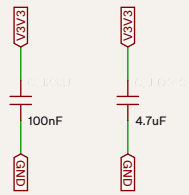
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VBUS pin references the USB protection diodes.

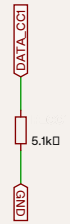
U_TEMP / BZ1 / D_STATUS / D_PWR

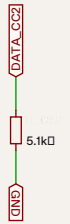
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PWM magnetic buzzer and RGB status need firmware.
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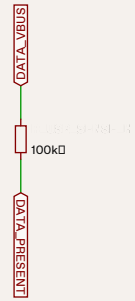
NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

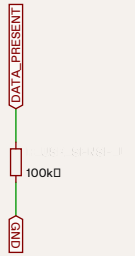
SHEET 4/13 - C_MCU, C_LOGIC, R_CC1, R_CC2, R_USB_SENSE_H, R_USB_SENSE_L











Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

COMPONENT PURPOSE

U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
Firmware must check PD before motor enable.
Internal clock avoids an external crystal.

U_PD - CH224K

USB-C PD sink requests a 15 V motor supply.
PG reports a completed power negotiation.

U_BUCK - AP63203WU-7

3.3 V buck for MCU, logic and PWM buzzer.
Diode OR accepts programming USB or PD.

U_DRV - A4988

Two current-regulated bridges drive the coils.
STEP/DIR interface; fixed 1/16 microstepping.
0.24 ohm sense resistors set phase current.

U_ESD - USBLC6-2SC6

Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the USB protection diodes.

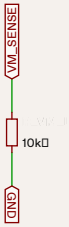
U_TEMP / BZ1 / D_STATUS / D_PWR

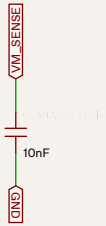
TMP102: I2C temperature, address 0x48.
PWM magnetic buzzer and RGB status need firmware.
Power LED indicates 3.3 V; RGB uses USB 5 V.

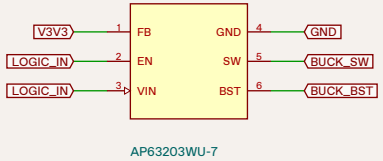
NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 5/13 - R_VM_H, R_VM_L, C_VM_SENSE, U_BUCK, D_PD, D_DATA













COMPONENT PURPOSE

U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
Firmware must check PD before motor enable.
Internal clock avoids an external crystal.

U_PD - CH224K

USB-C PD sink requests a 15 V motor supply.
PG reports a completed power negotiation.

U_BUCK - AP63203WU-7

3.3 V buck for MCU, logic and PWM buzzer.
Diode OR accepts programming USB or PD.

U_DRV - A4988

Two current-regulated bridges drive the coils.
STEP/DIR interface; fixed 1/16 microstepping.
0.24 ohm sense resistors set phase current.

U_ESD - USBLC6-2SC6

Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the USB protection diodes.

U_TEMP / BZ1 / D_STATUS / D_PWR

TMP102: I2C temperature, address 0x48.
PWM magnetic buzzer and RGB status need firmware.
Power LED indicates 3.3 V; RGB uses USB 5 V.

Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

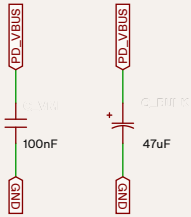
SHEET 6/13 - C_CP, C_VCP, C_VREG, C_DRV_LOGIC, C_VM, C_BULK











Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

COMPONENT PURPOSE

U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
Firmware must check PD before motor enable.
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U_PD - CH224K

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Two current-regulated bridges drive the coils.
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U_ESD - USBLC6-2SC6

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U_TEMP / BZ1 / D_STATUS / D_PWR

TMP102: I2C temperature, address 0x48.
PWM magnetic buzzer and RGB status need firmware.
Power LED indicates 3.3 V; RGB uses USB 5 V.

NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 7/13 - R_SA, R_SB, R_REF_H, R_REF_L, C_REF, R_ENABLE













COMPONENT PURPOSE

U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
Firmware must check PD before motor enable.
Internal clock avoids an external crystal.

U_PD - CH224K

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U_BUCK - AP63203WU-7

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STEP/DIR interface; fixed 1/16 microstepping.
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U_ESD - USBLC6-2SC6

Clamps ESD on USB D+ and D- at the data port.
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U_TEMP / BZ1 / D_STATUS / D_PWR

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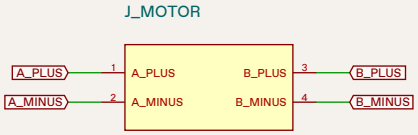
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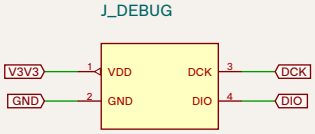
NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 8/13 - R_SLEEP, J_MOTOR, J_DEBUG, U_ESD, D_TVS, R_USB_BOOT

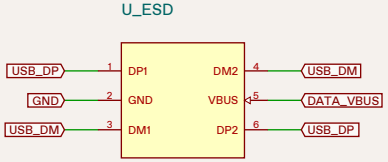




B4B-PH-K-S(LF)(SN)



PCB solder interface



USBLC6-2SC6





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PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

COMPONENT PURPOSE

U_MCU - CH32X035G8U6

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U_ESD - USBLC6-2SC6

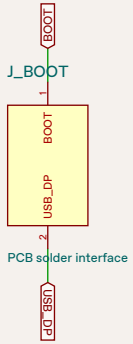
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VBUS pin references the USB protection diodes.

U_TEMP / BZ1 / D_STATUS / D_PWR

TMP102: I2C temperature, address 0x48.
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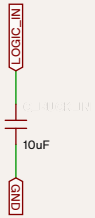
NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

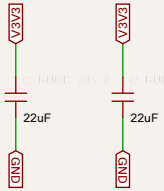
SHEET 9/13 - J_BOOT, L_BUCK, C_BOOTSTRAP, C_BUCK_IN, C_BUCK_OUT1, C_BUCK_OUT2











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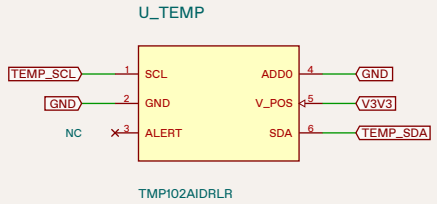
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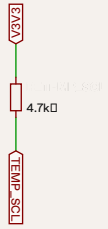
NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

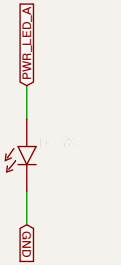
SHEET 10/13 - U_TEMP, C_TEMP, R_TEMP_SDA, R_TEMP_SCL, D_PWR, R_PWR_LED













COMPONENT PURPOSE

U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
Firmware must check PD before motor enable.
Internal clock avoids an external crystal.

U_PD - CH224K

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Clamps ESD on USB D+ and D- at the data port.
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U_TEMP / BZ1 / D_STATUS / D_PWR

TMP102: I2C temperature, address 0x48.
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Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 11/13 - D_STATUS, R_STATUS_R, Q_STATUS_R, R_STATUS_G, Q_STATUS_G, R_STATUS_B

D_STATUS



LTST-C19HE1WT



Q_STATUS_R



DTC114EU3HZGT106



Q_STATUS_G



DTC114EU3HZGT106



COMPONENT PURPOSE

U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
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NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

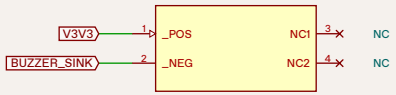
SHEET 12/13 - Q_STATUS_B, BZ1, Q_BUZZER, D_BUZZER, R_BUZZER_GATE, R_BUZZER_PD

Q_STATUS_B



DTC114EU3HZGT106

BZ1



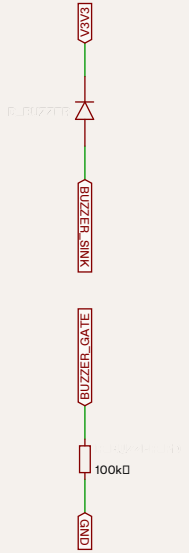
HYG-8503A

Q_BUZZER



AO3400A





Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

COMPONENT PURPOSE

U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
Firmware must check PD before motor enable.
Internal clock avoids an external crystal.

U_PD - CH224K

USB-C PD sink requests a 15 V motor supply.
PG reports a completed power negotiation.

U_BUCK - AP63203WU-7

3.3 V buck for MCU, logic and PWM buzzer.
Diode OR accepts programming USB or PD.

U_DRV - A4988

Two current-regulated bridges drive the coils.
STEP/DIR interface; fixed 1/16 microstepping.
0.24 ohm sense resistors set phase current.

U_ESD - USBLC6-2SC6

Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the USB protection diodes.

U_TEMP / BZ1 / D_STATUS / D_PWR

TMP102: I2C temperature, address 0x48.
PWM magnetic buzzer and RGB status need firmware.
Power LED indicates 3.3 V; RGB uses USB 5 V.

NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 13/13 - C_BUZZER



COMPONENT PURPOSE

U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
Firmware must check PD before motor enable.
Internal clock avoids an external crystal.

U_PD - CH224K

USB-C PD sink requests a 15 V motor supply.
PG reports a completed power negotiation.

U_BUCK - AP63203WU-7

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